

7 ~~3~~ Sep 2010

Separable enriched categories and absolute ends

An algebra A over a field is separable when there is an element $e = \sum_{i=1}^n a_i \otimes b_i \in A \otimes A$ such that

$$\sum_{i=1}^n a_i b_i = 1 \quad \text{and} \quad \sum_{i=1}^n (r a_i) \otimes b_i = \sum_{i=1}^n a_i \otimes (b_i r)$$

for all $a \in A$. It follows that A is finitely generated and projective as a left A -, right A -bimodule. There are many references.

An obvious generalization of this notion is as follows. Let $\mathcal{V}$ denote a monoidal category. A monoid A in $\mathcal{V}$ is separable when it is equipped with a morphism

$$\rho : I \longrightarrow A \otimes A$$

such that

$$\mu \circ \rho = \eta \quad \text{and} \quad (\mu \otimes 1)(1 \otimes \rho) = (1 \otimes \mu)(\rho \otimes 1).$$

Here I is the tensor unit object in $\mathcal{V}$ and $\eta : I \rightarrow A$, $\mu : A \otimes A \rightarrow A$ are the unit and multiplication for the monoid A . Using the string notation of [GTC I], we denote η , μ , ρ , respectively, as follows:

(2)

$!$, γ , $\cap$.

The axioms for separability then become

$$\text{cup} = ! \quad \text{and} \quad \text{cap} = \cap.$$

We wish to generalize this even further, still working with a monoidal category $\mathcal{V}$.

A $\mathcal{V}$ -category $\mathcal{A}$ is separable when it is non-empty and equipped with a family of morphisms

$$\rho : I \longrightarrow \mathcal{A}(B, A) \otimes \mathcal{A}(A, B), \quad A, B \in \mathcal{A},$$

such that

$$(1) \quad \begin{array}{ccc} I & \xrightarrow{\rho} & \mathcal{A}(B, A) \otimes \mathcal{A}(A, B) \\ & \searrow \eta & \downarrow \mu \\ & & \mathcal{A}(A, A) \end{array} \quad \text{and}$$

$$(2) \quad \begin{array}{ccc} \mathcal{A}(A, C) & \xrightarrow{1 \otimes \rho} & \mathcal{A}(A, C) \otimes \mathcal{A}(B, A) \otimes \mathcal{A}(A, B) \\ \rho \otimes 1 \downarrow & & \downarrow \mu \otimes 1 \\ \mathcal{A}(B, C) \otimes \mathcal{A}(C, B) \otimes \mathcal{A}(A, C) & \xrightarrow{1 \otimes \mu} & \mathcal{A}(B, C) \otimes \mathcal{A}(A, B). \end{array}$$

Now η and μ denote identities and compositions in the $\mathcal{V}$ -category $\mathcal{A}$. The diagonal of the

(3)

square (2) will be denoted by

$$\delta : A(A, C) \longrightarrow A(B, C) \otimes A(A, B);$$

it satisfies the Frobenius identity:

$$(2) \quad \begin{array}{ccc} A(C, D) \otimes A(A, C) & \xrightarrow{1 \otimes \delta} & A(C, D) \otimes A(B, C) \otimes A(A, B) \\ \delta \otimes 1 \downarrow & & \downarrow \mu \otimes 1 \\ A(B, D) \otimes A(C, B) \otimes A(A, C) & \xrightarrow{1 \otimes \mu} & A(B, D) \otimes A(A, B). \end{array}$$

We recover ρ from δ via $\rho = \delta \circ \eta$ where $C = A$.

We can adapt string notation to work in this situation. For the morphisms

$$\eta : I \longrightarrow A(A, A),$$

$$\mu : A(B, C) \otimes A(A, B) \longrightarrow A(A, C),$$

$$\delta : A(A, C) \longrightarrow A(B, C) \otimes A(A, B), \text{ and}$$

$$\rho : I \longrightarrow A(B, A) \otimes A(A, B),$$

we respectively write

$$A \circ A, \quad \begin{array}{c} B \\ \diagup \quad \diagdown \\ A \quad C \end{array}, \quad \begin{array}{c} A \quad C \\ \diagdown \quad \diagup \\ B \end{array}, \quad \begin{array}{c} A \quad B \quad A \\ \text{curved line} \end{array}$$

Lemma 1 For a separable V -category A , the following diagram commutes.

$$\begin{array}{ccc}
 I & \xrightarrow{\rho} & A(B, A) \otimes A(A, B) \xrightarrow{1 \otimes \rho \otimes 1} A(B, A) \otimes A(C, B) \otimes A(B, C) \otimes A(A, B) \\
 & \searrow \rho & \downarrow \mu \otimes \mu \\
 & & A(C, A) \otimes A(A, C)
 \end{array}$$

In diagrams,

$$\begin{array}{c} B \\ \cup \\ A \end{array} \begin{array}{c} \cup \\ C \end{array} = \begin{array}{c} \cup \\ C \end{array}$$

Proof

$$\begin{array}{c} B \\ \cup \\ A \end{array} \begin{array}{c} \cup \\ C \end{array} = \begin{array}{c} A \quad B \\ \cup \quad \cup \\ C \end{array} = \begin{array}{c} A \quad B \\ \cup \quad \cup \\ C \end{array} = \begin{array}{c} A \quad \circ \\ \cup \\ C \end{array} = \begin{array}{c} \cup \\ C \end{array} = \begin{array}{c} A \quad \circ \\ \cup \\ C \end{array} = \begin{array}{c} \cup \\ C \end{array} \quad \square$$

For a V -category A , a right A -module M consists of a family of objects MA , $A \in A$, in V equipped with morphisms

$$\mu_r : MB \otimes A(A, B) \longrightarrow MA$$

satisfying the conditions for an action. For μ_r we write

$$\begin{array}{c} B \\ \cup \\ M \quad A \end{array}$$

and the action conditions become

(5)

$$\begin{array}{c} \diagup \\ A \\ \diagdown \end{array} \begin{array}{c} \diagdown \\ A \\ \diagup \end{array} = \begin{array}{c} | \\ A \end{array} \quad \text{and} \quad \begin{array}{c} \diagup \\ C \\ \diagdown \end{array} \begin{array}{c} \diagdown \\ B \\ \diagup \end{array} = \begin{array}{c} \diagup \\ C \\ \diagdown \end{array} \begin{array}{c} \diagdown \\ B \\ \diagup \end{array}$$

Left A -module N is defined symmetrically:

$$\mu_l : A(A, B) \otimes N A \longrightarrow N B, \quad \begin{array}{c} \diagup \\ B \\ \diagdown \end{array} \begin{array}{c} \diagdown \\ A \\ \diagup \end{array} N.$$

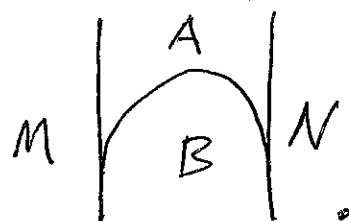
Suppose M and N are right and left A -modules, respectively. Define morphisms

$$e_{A,B} : M A \otimes N A \longrightarrow M B \otimes N B$$

to be the composite

$$\begin{array}{ccc}
 M A \otimes N A & \xrightarrow{1 \otimes \rho \otimes 1} & M A \otimes A(B, A) \otimes A(A, B) \otimes N A \\
 & & \searrow \mu_r \otimes \mu_l \\
 & & M B \otimes N B
 \end{array}$$

when A is separable. Using strings, we draw e_{AB} as

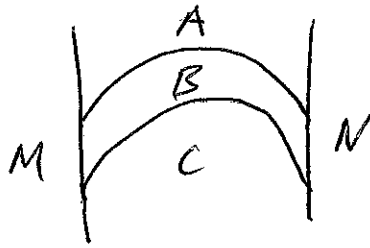


Proposition 2 The following diagram commutes.

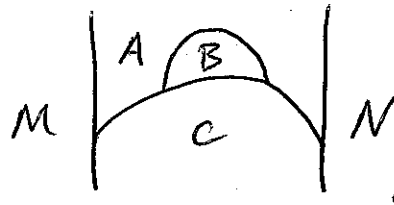
$$\begin{array}{ccc}
 M A \otimes N A & \xrightarrow{e_{AC}} & M C \otimes N C \\
 \searrow e_{AB} & & \nearrow e_{BC} \\
 & M B \otimes N B &
 \end{array}$$

(6)

Proof The lower path in the triangle is



which, by the action conditions, is



Lemma 1 now gives the result. $\square$

The chaotic category on a set K is denoted by K_{ch} : its objects are the elements of K and its homs $K_{ch}(i, j)$ are all singletons.

A clique^[GTCL] in a category $\mathcal{K}$ is a functor $K_{ch} \rightarrow \mathcal{K}$ from a non-empty chaotic category.

We write $Q\mathcal{K}$ for the idempotent splitting completion of the ordinary category $\mathcal{K}$. Objects are pairs (X, e) where $e: X \rightarrow X$ is an idempotent $e^2 = e$ in $\mathcal{K}$. Morphisms $f: (X, e) \rightarrow (X', e')$ are morphisms $f: X \rightarrow X'$ in $\mathcal{K}$ satisfying $f = fe = e'f$. We identify $X \in \mathcal{K}$ with $(X, 1_X) \in Q\mathcal{K}$.

Corollary 3 Each e_{AA} is an idempotent on $MA \otimes NA$ and the diagram (7)

$$e_{AB} : (MA \otimes NA, e_{AA}) \longrightarrow (MB \otimes NB, e_{BB}),$$

$A, B \in \mathcal{A},$

is a clique in $\mathcal{QV}_0$. $\square$

For a clique $X : K_{ch} \rightarrow \mathcal{E}$, since every object of K_{ch} is both initial and terminal, every value $X(i)$ of X is both a limit and colimit for the functor X .

Corollary 4 If idempotents split in $\mathcal{V}_0$ then the diagram

$$(4) \quad MA \otimes NA \xrightarrow{e_{AB}} MB \otimes NB, \quad A, B \in \mathcal{A},$$

has an absolute colimit (and limit) in $\mathcal{V}_0$. $\square$

To be specific, we have a cocone

$$\begin{array}{ccc} MA \otimes NA & \xrightarrow{e_{AB}} & MB \otimes NB \\ & \searrow h_A \quad \swarrow h_B & \\ & T & \end{array}$$

and morphisms $k_A : T \longrightarrow MA \otimes NA$ satisfying

$$e_{AB} = k_B h_A \quad \text{and} \quad h_A k_A = 1_T.$$

(8)

From these equations it can be deduced that $(h_A)_{A \in A}$ is a colimit which, consequently, is absolute. Also $(k_A)_{A \in A}$ is a limit cone.

{ There is no Proposition 5. }

that the cocore is a colimit. Hence such colimits are preserved by all functors. $\square$

Recall that the tensor product (or composite) of a right A -module M and a left A -module N is defined to be the colimit

$$(5) \quad \begin{array}{ccc} MB \otimes_A (A, B) \otimes_A NA & \xrightarrow{\mu_r \otimes 1} & MA \otimes NA \\ \downarrow 1 \otimes \mu_l & & \downarrow p_A \\ MB \otimes NB & \xrightarrow[p_B]{} & M \otimes_A N \end{array}$$

in $\mathcal{V}$, when it exists.

Proposition 6 A family of morphisms

$f_A : MA \otimes NA \longrightarrow X, \quad A \in A,$
is a cocone for the solid diagram of (5)
if and only if it is a cocone for (4).

Proof A cocone for (5) means

$$(6) \quad \begin{array}{c} \text{Diagram 1: } M \text{ and } N \text{ are inputs to a node } \otimes. \text{ The left input } M \text{ is labeled } B, \text{ and the right input } N \text{ is labeled } A. \\ \text{Diagram 2: } M \text{ and } N \text{ are inputs to a node } \otimes. \text{ The left input } M \text{ is labeled } B, \text{ and the right input } N \text{ is labeled } A. \end{array} =$$

A monoidal functor $F: \mathcal{V} \rightarrow \mathcal{W}$ will be ⁽¹¹⁾
called separable when there is a natural family
of morphisms

$$\delta_{u,v} : F(u \otimes v) \longrightarrow Fu \otimes Fv$$

such that

$$(8) \quad \begin{array}{ccc} F(u \otimes v) & \xrightarrow{\delta_{u,v}} & Fu \otimes Fv \\ & \searrow 1 & \downarrow \varphi_{u,v} \\ & & F(u \otimes v), \end{array}$$

$$(9) \quad \begin{array}{ccc} F(u \otimes v) \otimes Fw & \xrightarrow{\delta_{u,v} \otimes 1} & Fu \otimes Fv \otimes Fw \\ \varphi_{u \otimes v, w} \downarrow & & \downarrow 1 \otimes \varphi_{v,w} \\ F(u \otimes v \otimes w) & \xrightarrow{\delta_{u,v \otimes w}} & Fu \otimes F(v \otimes w) \end{array}$$

$$(10) \quad \begin{array}{ccc} Fu \otimes F(v \otimes w) & \xrightarrow{1 \otimes \delta_{v,w}} & Fu \otimes Fv \otimes Fw \\ \varphi_{u, v \otimes w} \downarrow & & \downarrow \varphi_{u,v} \otimes 1 \\ F(u \otimes v \otimes w) & \xrightarrow{\delta_{u \otimes v, w}} & F(u \otimes v) \otimes Fw \end{array}$$

where $\varphi_{u,v}$ and $\varphi_0 : I \rightarrow FI$ provide the monoidal
structure on F , and we have suppressed associat-
ivity constraints.

Eilenberg-Kelly [CC] showed that a monoidal ⁽¹²⁾ functor $F: \mathcal{V} \rightarrow \mathcal{W}$ takes a $\mathcal{V}$ -category A to a $\mathcal{W}$ -category $F_* A$ with the same objects and

$$(F_* A)(A, B) = F(A(A, B)).$$

Proposition 8 If $F: \mathcal{V} \rightarrow \mathcal{W}$ is a separable monoidal
functor and A is a separable $\mathcal{V}$ -category then
 $F_* A$ is a separable $\mathcal{W}$ -category.

Proof Define ρ for $F_* A$ to be the composite

$$I \xrightarrow{\varphi_0} FI \xrightarrow{F\rho} F(A(B, A) \otimes A(A, B)) \xrightarrow{\delta} FA(B, A) \otimes FA(A, B).$$

To save space we write (AB) for $A(A, B)$. The following proves (1) for this ρ :

$$\begin{array}{ccccc} I & \xrightarrow{\varphi_0} & FI & \xrightarrow{F\rho} & F((BA) \otimes (AB)) & \xrightarrow{\delta} & F(BA) \otimes F(AB) \\ & & & & \searrow \text{(8)} & \downarrow \varphi & \\ & & & & & & F((BA) \otimes (AB)) \\ & & & & \searrow \text{(1)} & \downarrow F\mu & \\ & & & & & & F(AA) \end{array}$$

η (from I to $F(AA)$)
 $F\eta$ (from FI to $F(AA)$)

The following proves (2) for $F_* A$:

$$\begin{array}{c}
F(AC) \xrightarrow{1 \otimes \varphi_0} F(AC) \otimes FI \xrightarrow{1 \otimes \varphi_P} F(AC) \otimes F((BA) \otimes (AB)) \\
\downarrow \varphi_0 \otimes 1 \quad \searrow 1 \quad \downarrow \varphi \quad \text{nat.} \quad \searrow \varphi \quad \downarrow 1 \otimes \delta \\
FI \otimes F(AC) \xrightarrow{\varphi} F(AC) \xrightarrow{F(1 \otimes \varphi)} F((AC) \otimes (BA) \otimes (AB)) \quad (10) \quad \downarrow \varphi_2 \otimes 1 \\
\downarrow F\varphi \otimes 1 \quad \text{nat.} \quad \downarrow F(\varphi \otimes 1) \quad \downarrow F(\mu \otimes 1) \quad \delta \quad \downarrow \text{nat.} \quad \downarrow F\mu \otimes 1 \\
F((BC) \otimes (CB)) \otimes F(AC) \xrightarrow{\varphi} F((BC) \otimes (CB) \otimes (AC)) \xrightarrow{F(1 \otimes \mu)} F((BC) \otimes (AB)) \quad \downarrow \delta \quad \downarrow \delta \\
\downarrow \delta \otimes 1 \quad (9) \quad \downarrow \delta \quad \text{nat.} \quad \downarrow \delta \quad \downarrow \delta \\
F(BC) \otimes F(CB) \otimes F(AC) \xrightarrow{1 \otimes \varphi_2} F(BC) \otimes F((CB) \otimes (AC)) \xrightarrow{1 \otimes F\mu} F(BC) \otimes F(AB)
\end{array}$$

□

Each right A -module M determines a right F_*A -module F_*M defined by

$$(F_*M)A = FMA$$

$\mu_r^F : FM \otimes FA(A, B) \xrightarrow{\varphi} F(MB \otimes A(A, B)) \xrightarrow{F\mu_r} FMA$,
and similarly for left A -modules. If A is separable, by Proposition 8 we have the family

$$FMA \otimes FNA \xrightarrow{e_{AB}^F} FMB \otimes FNB$$

of Proposition 2 for the $\mathcal{W}$ -category F_*A and modules F_*M and F_*N .

Proposition 9 The following diagram commutes

$$(II) \quad \begin{array}{ccc} FMA \otimes FNA & \xrightarrow{e_{AB}^F} & FMB \otimes FNB \\ \varphi \downarrow & & \downarrow \varphi \\ F(MA \otimes NA) & \xrightarrow{Fe_{AB}} & F(MB \otimes NB). \end{array}$$

Proof

$$\begin{array}{c} \begin{array}{ccccc} & & FMA \otimes F(BA) \otimes (AB) \otimes FNA & & \\ & \nearrow 1 \otimes F \rho \otimes 1 & & \nwarrow 1 \otimes \delta \otimes 1 & \\ & FMA \otimes FI \otimes FNA & & FMA \otimes F(BA) \otimes F(AB) \otimes FNA & \\ & \nearrow 1 \otimes \varphi_0 \otimes 1 & \searrow 1 \otimes F \rho \otimes 1 & \nearrow 1 \otimes \varphi \otimes 1 & \\ FMA \otimes FNA & & FMA \otimes F(BA) \otimes (AB) \otimes FNA & & \\ \varphi \downarrow & \nearrow \varphi_3 & \downarrow \varphi_3 & \nwarrow \varphi & \downarrow \varphi \otimes \varphi \\ F(MA \otimes NA) & \xrightarrow{F(1 \otimes \rho \otimes 1)} & F(MA \otimes (BA) \otimes (AB) \otimes NA) & \xleftarrow{\varphi} & F(MA \otimes (BA)) \otimes F((AB) \otimes NA) \\ & \searrow Fe & \downarrow F(\mu_1 \otimes \mu_2) & \nearrow \varphi & \swarrow F\mu_1 \otimes F\mu_2 \\ & & F(MB \otimes NB) & & FMB \otimes FNB \end{array} \\ & & & & \\ & & & & \square \end{array}$$

It follows from Corollary 4 and Proposition 9 that φ induces a morphism

$$\bar{\varphi} : F_* M \otimes_{F_* A} F_* N \longrightarrow F(M \otimes_A N)$$

on the (absolute) colimits of the rows of (II). Notice that, if $F: \mathcal{V} \rightarrow \mathcal{W}$ is strong monoidal then $\bar{\varphi}$ is invertible.

Suppose N and L are both left A -modules.
A functor $\text{hom}_A(N, L): \mathcal{V}^{\mathcal{V}} \rightarrow \text{Set}$ is defined pointwise as the limit

$$\begin{array}{ccc}
 \text{hom}_A(N, L)(X) & \xrightarrow{\quad p_A \quad} & \mathcal{V}(NA \otimes X, LA) \\
 \downarrow p_B & & \downarrow A(A, B) \otimes - \\
 & & \mathcal{V}(A(A, B) \otimes NA \otimes X, A(A, B) \otimes LA) \\
 & & \downarrow \mathcal{V}(1, \mu_L) \\
 \mathcal{V}(NB \otimes X, LB) & \xrightarrow[\mathcal{V}(\mu_L \otimes 1, 1)]{} & \mathcal{V}(A(A, B) \otimes NA \otimes X, LB).
 \end{array}$$

A (right) hom for N and L over A is a representing object $\underline{H}_A(N, L)$ for $\text{hom}_A(N, L)$, when it exists.

If A is separable we can define a natural family of functions

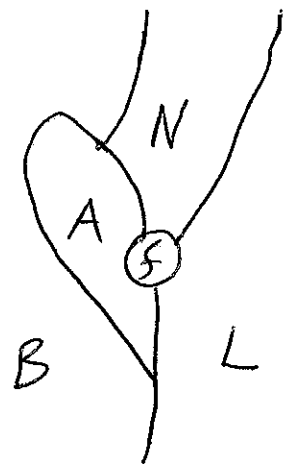
$$\underline{\eta}_{A, B, X}: \mathcal{V}(NA \otimes X, LA) \longrightarrow \mathcal{V}(NB \otimes X, LB)$$

taking $f: NA \otimes X \rightarrow LA$ to the composite

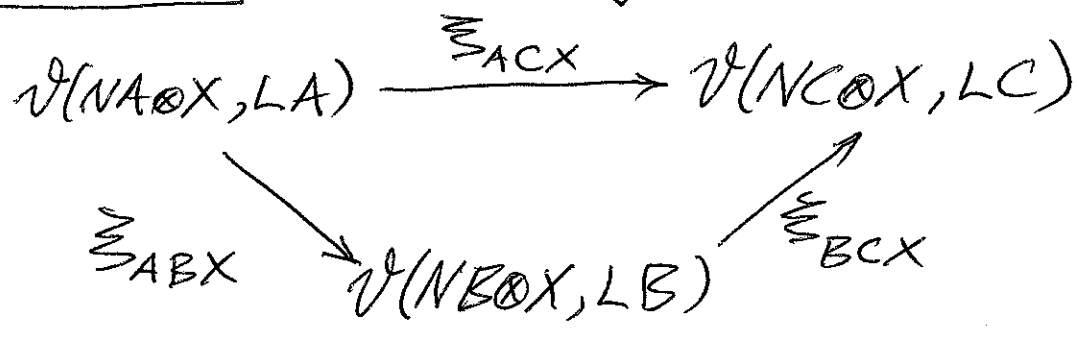
$$\begin{aligned}
 NB \otimes X &\xrightarrow{p \otimes 1 \otimes 1} A(A, B) \otimes A(B, A) \otimes NB \otimes X \xrightarrow{1 \otimes \mu_L \otimes 1} A(A, B) \otimes NA \otimes X \\
 &\xrightarrow{1 \otimes f} A(A, B) \otimes LA \xrightarrow{\mu_B} LB.
 \end{aligned}$$

In string diagrams:

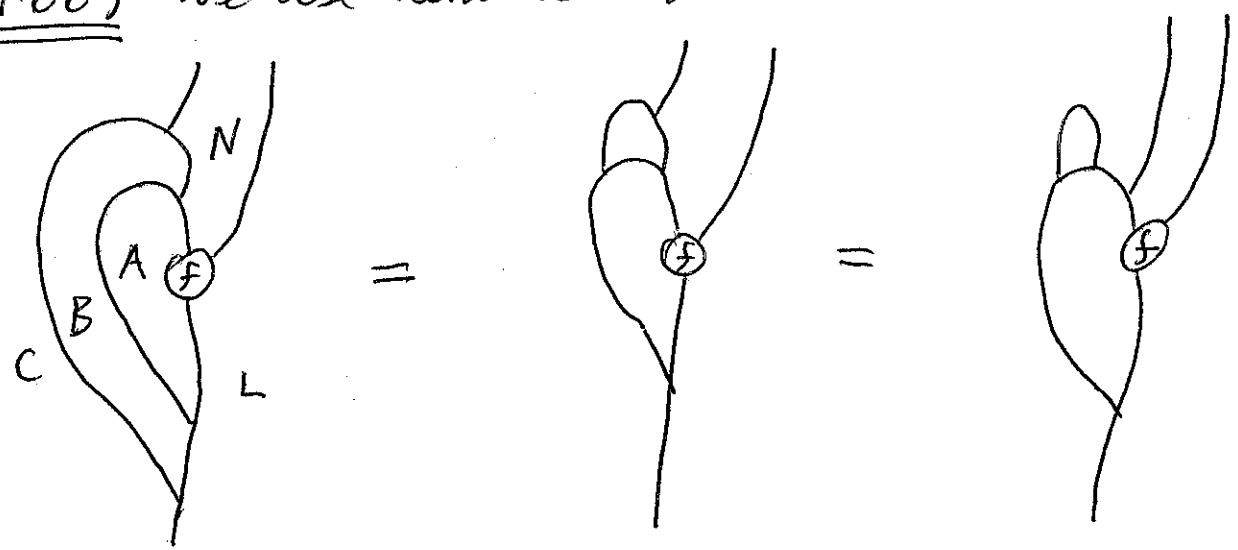
$\Sigma(f) =$



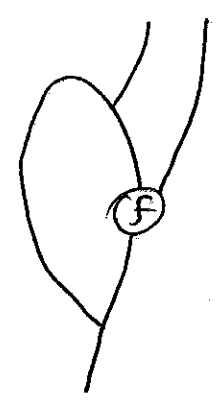
Proposition 10 The following triangle commutes.



Proof We use Lemma 1:



(Lem 1)
=



□

Proposition 11 The limit in Set of the diagram

$$\mathcal{V}(NA \otimes X, LA) \xrightarrow{\sum_{A \in \mathcal{A}} \mathbf{f}_A} \mathcal{V}(NB \otimes X, LB), \quad A, B \in \mathcal{A},$$

is $\text{hom}_{\mathcal{A}}(N, L)(X)$.

Proof We need to see that a family of morphisms

$$\mathbf{f}_A : NA \otimes X \longrightarrow LA, \quad A \in \mathcal{A},$$

satisfies

$$(12) \quad \begin{array}{c} N \\ \swarrow \quad \searrow \\ A \quad B \\ \searrow \quad \swarrow \\ \textcircled{f} \\ \downarrow \\ L \end{array} = \begin{array}{c} A \quad N \\ \searrow \quad \swarrow \\ B \quad \textcircled{f} \\ \downarrow \\ L \end{array}$$

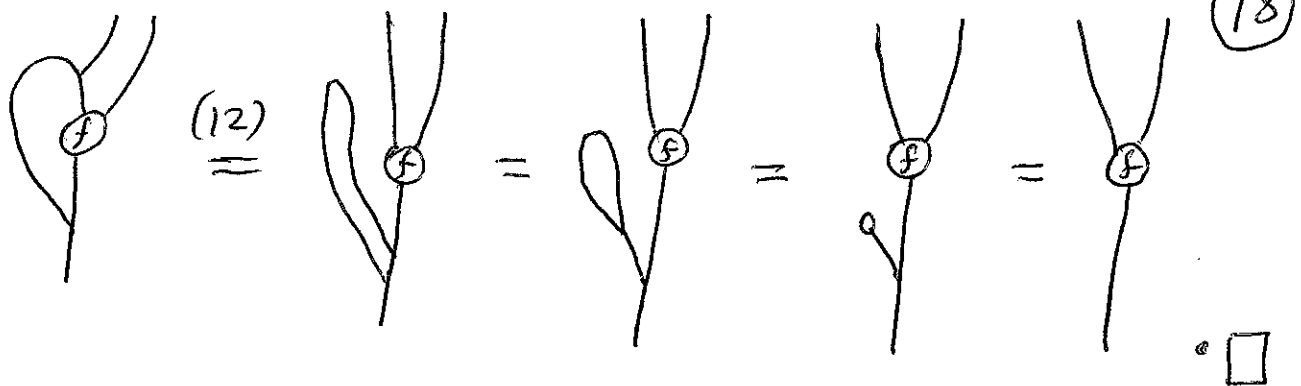
if and only it satisfies

$$(13) \quad \begin{array}{c} N \\ \swarrow \quad \searrow \\ A \quad B \\ \searrow \quad \swarrow \\ \textcircled{f} \\ \downarrow \\ L \end{array} = \begin{array}{c} N \\ \searrow \quad \swarrow \\ \textcircled{f} \\ \downarrow \\ L \end{array}.$$

Assuming (13), we have

$$\begin{array}{c} N \\ \searrow \quad \swarrow \\ \textcircled{f} \\ \downarrow \\ L \end{array} \stackrel{(13)}{=} \begin{array}{c} N \\ \swarrow \quad \searrow \\ A \quad B \\ \searrow \quad \swarrow \\ \textcircled{f} \\ \downarrow \\ L \end{array} = \begin{array}{c} Y \\ \swarrow \quad \searrow \\ \textcircled{f} \\ \downarrow \\ L \end{array} \stackrel{(13)}{=} \begin{array}{c} Y \\ \searrow \quad \swarrow \\ \textcircled{f} \\ \downarrow \\ L \end{array}.$$

Assuming (12), we have



A right hom of object Y and Z in $\mathcal{V}$ will be denoted by Z^Y :

$$\mathcal{V}(X, Z^Y) \cong \mathcal{V}(Y \otimes X, Z).$$

Corollary 12 If idempotents split in $\mathcal{V}$ and $\mathcal{V}$ is right closed then, for a separable $\mathcal{V}$ -category $\mathcal{A}$, all left $\mathcal{A}$ -modules N and L have a right hom
 $H_{\mathcal{A}}(N, L).$

Proof The functions $\Xi_{A,B,X}$ transport to a natural family

$$\mathcal{V}(X, (LA)^{(NA)}) \longrightarrow \mathcal{V}(X, (LB)^{(NB)})$$

which, by Yoneda's Lemma, has the form $\mathcal{V}(1, t_{AB})$ for a family of morphisms

$$(14) \quad t_{A,B} : (LA)^{(NA)} \longrightarrow (LB)^{(NB)}.$$

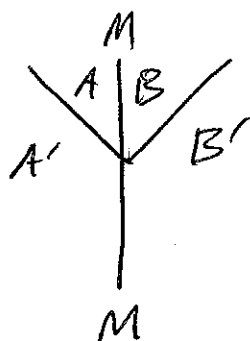
By Proposition 10 these form a clique in $\mathcal{Q}\mathcal{V}_0 \cong \mathcal{V}_0$. Hence the diagram (14) has an (absolute) limit $H_{\mathcal{A}}(N, L)$ which, by Proposition 11, represents the functor $\text{hom}_{\mathcal{A}}(N, L).$ $\square$

For any monoidal category $\mathcal{V}$, a module $M: \mathcal{A} \rightarrow \mathcal{B}$ for $\mathcal{V}$ -categories $\mathcal{A}$ and $\mathcal{B}$ is a family of objects

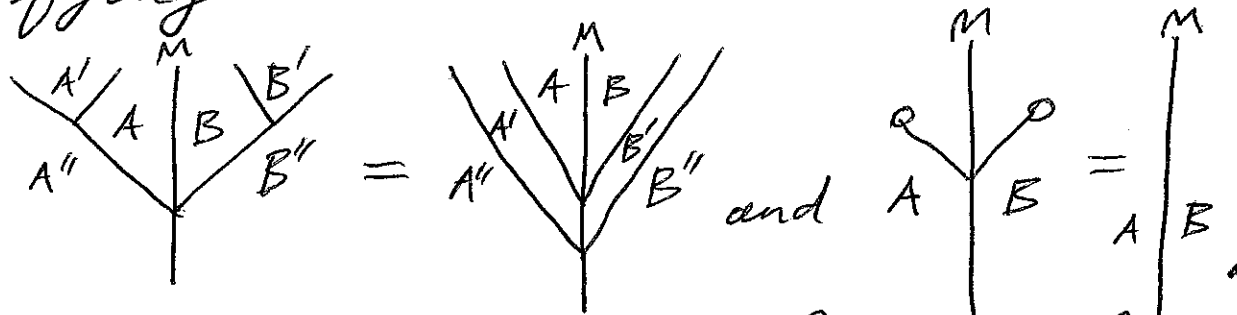
$M(B, A) \in \mathcal{V}$ for $A \in \mathcal{A}, B \in \mathcal{B}$ equipped with a family of action morphisms

$$\mu: \mathcal{A}(A, A') \otimes M(B, A) \otimes \mathcal{B}(B', B) \rightarrow M(B', A')$$

denoted by



satisfying



A morphism $f: M \rightarrow N: \mathcal{A} \rightarrow \mathcal{B}$ of modules is a family of morphisms

$$f_{BA}: M(B, A) \rightarrow N(B, A), \quad A \in \mathcal{A}, B \in \mathcal{B},$$

preserving the actions. It is easy to see that when we have modules

$$\mathcal{A} \xrightarrow{M} \mathcal{B} \xrightarrow{N} \mathcal{C},$$

(20)

if $M \otimes_B N$ exists (see (5)), actions are induced in the obvious way to obtain a module

$$M \otimes_B N : A \rightarrow C.$$

Proposition 13 There is a bicategory $V\text{-Mod}_{\text{sep}}$ when idempotents split in V . The objects are separable V -categories. The morphisms are modules. The 2-cells are module morphisms.
Composition of modules $M : A \rightarrow B$, $N : B \rightarrow C$ is defined by

$$N \circ M = M \otimes_B N : A \rightarrow C.$$

The identity module of A is the family of
homs $A(A', A)$ with composition as action.
Associativity constraints are induced by
those of V . Unit constraints are induced by
actions.

Proof Use Corollary 7. $\square$

Proposition 14 In Proposition 13, if V is also
right closed, all right extensions exist
in the bicategory $V\text{-Mod}_{\text{sep}}$.

Proof Use Corollary 12. $\square$

For ordinary categories, separability is not very interesting however.

Proposition 15 For $V = \underline{\text{Set}}$, a V -category A is separable if and only if it is equivalent to 1.

Proof The function $p: 1 \rightarrow A(B, A) \times A(A, B)$ picks out a pair of morphisms

$$a_B^A: B \rightarrow A \quad \text{and} \quad b_A^B: A \rightarrow B$$

such that $a_B^A b_A^B = 1$ and, for all $f: A \rightarrow C$,

$$f a_B^A = a_B^C \quad \text{and} \quad b_A^B = b_C^B f.$$

It follows that

$$f = a_B^C b_A^B$$

for any choice of B . So each $A(A, C)$ is a singleton. $\square$

What we have looked at so far has required no braiding on $\mathcal{V}$, and no completeness or cocompleteness apart from splitting idempotents. Now we go to the other extreme and assume $\mathcal{V}$ is symmetric closed monoidal, complete and cocomplete. Then we can take advantage of the enriched category theory of [Kelly's book]. In particular, we have the $\mathcal{V}$ -functor category $[A^{\text{op}} \otimes A, \mathcal{V}]$ for each small $\mathcal{V}$ -category A and we know that it is the free cocompletion of $A^{\text{op}} \otimes A$.

An object P of a $\mathcal{V}$ -category $\mathcal{K}$ is called small projective when $\mathcal{K}(P, -) : \mathcal{K} \rightarrow \mathcal{V}$ preserves small colimits.

Borceux-Vitale [Azumaya categories] define a $\mathcal{V}$ -category A to be separable when the hom- $\mathcal{V}$ -functor $A(-, -)$ is a small projective object of $[A^{\text{op}} \otimes A, \mathcal{V}]$. We shall say A is BV-separable in this case to distinguish it from our earlier notion.

Proposition 16 For $\mathcal{V}$ the monoidal category of abelian groups, a one-object $\mathcal{V}$ -category A is separable if and only if it is BV-separable.

Proof Proposition 1.1 of [BV:alg.cat]. $\square$

Proposition 17 The $\mathcal{V}$ -functor

$[A^{\text{op}} \otimes A, \mathcal{V}](A(-, -), -) : [A^{\text{op}} \otimes A, \mathcal{V}] \rightarrow \mathcal{V}$
is isomorphic to

$$\int_A : [A^{\text{op}} \otimes A, \mathcal{V}] \rightarrow \mathcal{V}$$

which takes each $M : A^{\text{op}} \otimes A \rightarrow \mathcal{V}$ to its end $\int_A M(A, A)$.

Proof We have the calculation:

$$\begin{aligned} [A^{\text{op}} \otimes A, \mathcal{V}](A(-, -), M) &= \int_{A, B} [A(A, B), M(A, B)] \\ &\cong \int_A M(A, A) \end{aligned}$$

which uses the "end form" of the Yoneda Lemma. $\square$

Corollary 18 A V -category A is BV-separable if and only if $\int_A : [A^o \otimes A, V] \rightarrow V$ preserves small colimits.

The composite of $\int_A$ with the Yoneda embedding is the V -functor

$$W_A : A^o \otimes A \longrightarrow V$$

defined by

$$W_A(C, D) = \int_A A(A, D) \otimes A(C, A).$$

By the general theory of cocompletion, it follows from Corollary 18, if A is BV-separable, the end $\int_A$ is the weighted colimit

$$\operatorname{colim}(W_A, -)$$

and has right adjoint

$$V \longrightarrow [A^o \otimes A, V]$$

$$X \longmapsto (C, D) \longmapsto \int_A [X, A(A, D) \otimes A(C, A)]$$

Recall that a module $M : I \rightarrow B$ has a right adjoint in the bicategory $\mathcal{V}\text{-Mod}$ if and only if $M \in [B^{\text{op}}, \mathcal{V}]$ is small projective. Similarly, a module $N : \mathcal{C} \rightarrow I$ has a left adjoint if and only if $N \in [\mathcal{C}, \mathcal{V}]$ is small projective (since it is the same as $N^{\text{op}} : I \rightarrow \mathcal{C}^{\text{op}}$ having a right adjoint).

Recall also that we have modules

$$E_A : A^{\text{op}} \otimes A \rightarrow I \quad \text{and} \quad N_A : I \rightarrow A \otimes A^{\text{op}},$$

for any $\mathcal{V}$ -category A , defined by

$$E_A(*, A, B) = A(A, B) = N_A(B, A, *).$$

(In fact these are counit and unit for a biduality $A^{\text{op}} \dashv_{\pm} A$ in the monoidal bicategory $\mathcal{V}\text{-Mod}$.)

Proposition 19 The following conditions on a $\mathcal{V}$ -category A are equivalent:

- (i) A is BV -separable;
- (ii) $E_A : A^{\text{op}} \otimes A \rightarrow I$ has a left adjoint module;
and
- (iii) $N_A : I \rightarrow A \otimes A^{\text{op}}$ has a right adjoint module.

Proof Under the equivalences

$$[A^{\text{op}} \otimes A, \mathcal{V}] \simeq \mathcal{V}\text{-Mod}(A^{\text{op}} \otimes A, I) \simeq \mathcal{V}\text{-Mod}(I, A \otimes A^{\text{op}})$$

the hom $A(-, -)$ is taken to E_A which is then taken to N_A . If one is small projective, all are. $\square$

Proposition 20 If A is BV-separable then
end

$$\int_A : [A^{\text{op}} \otimes A, \mathcal{X}] \longrightarrow \mathcal{X}$$

and coend

$$\int^A : [A^{\text{op}} \otimes A, \mathcal{X}] \longrightarrow \mathcal{X}$$

are absolute (weighted) limits and colimits,
respectively. They exist for any Cauchy
complete $\mathcal{V}$ -category $\mathcal{X}$.

Proof We have

$$\begin{aligned} \lim(E_A, S) &= \int_{A, B} [A(A, B), S(A, B)] \\ &\simeq \int_A S(A, A) \quad \text{and} \end{aligned}$$

$$\text{colim}(N_A, S) = \int^{A, B} A(B, A) \otimes S(A, B) \\ \simeq \int^A S(A, A),$$

(27)

showing that end is limit weighted by E_A while coend is colimit weighted by N_A .

By [StAcieec], $\text{lim}(E_A, -)$ is absolute since E_A has a left adjoint and $\text{colim}(N_A, -)$ is absolute since N_A has a right adjoint (i.e. is a Cauchy module). $\square$

given modules as in the triangle

$$\begin{array}{ccc} A & \xrightarrow{M} & B \\ & \searrow L & \swarrow N \\ & C & \end{array},$$

we have

$$H_{\sim A}(M, L) = \int_A [M(-, A), L(\sim, A)]$$

$$N \circ M = M \otimes_B N = \int^B M(B, -) \otimes N(\sim, B)$$

$$H_{\sim C}^{\circ}(N, L) = \int_C [N(C, -), L(C, \sim)]$$

involving absolute ends and coends when A, B, C are BV-separable, respectively.